

How AI Is Built and Implemented

The lifecycle, the mechanism inside the model, the scale economics, and the software layer that turns a model into a product.

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Confidence-tagged · provenance at end

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1. The lifecycle: two phases with different economics

"AI" as shipped today is a large neural network whose behavior is set in a one-time **training** phase and then used in a per-request **inference** phase. Training produces a frozen file of numbers called weights. Inference is that file being run, over and over, for every request. Every later cost and performance question traces back to this split: building is a capital expense paid once, implementing is an operating expense paid per token.

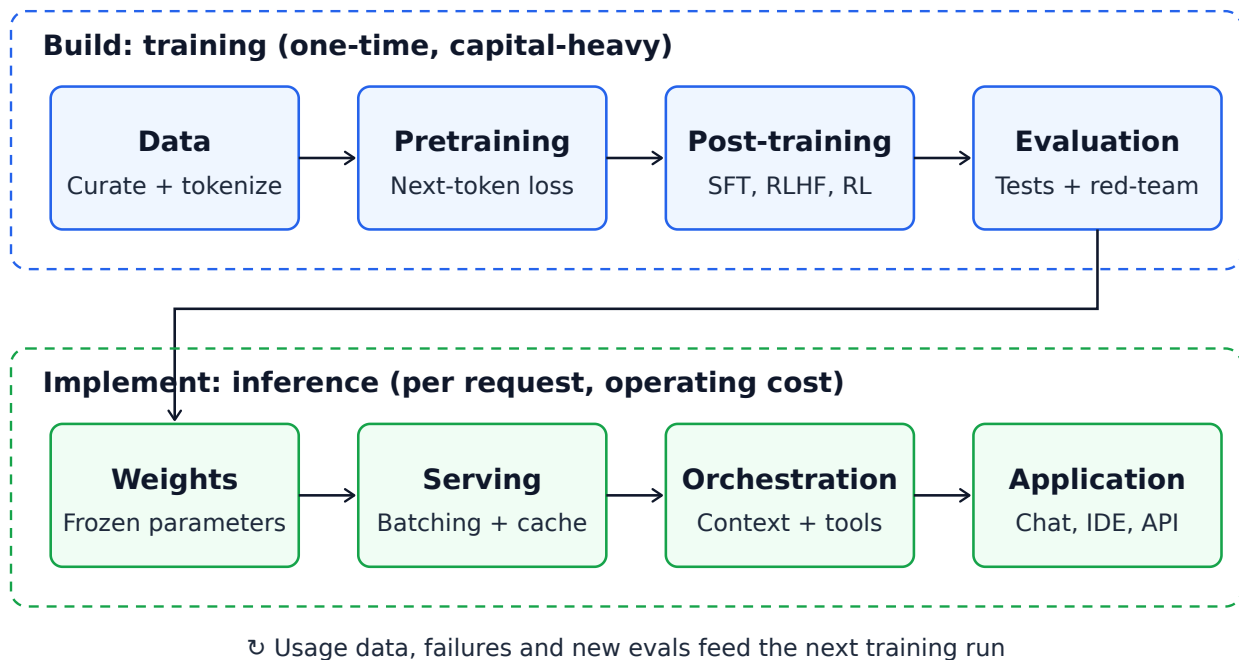


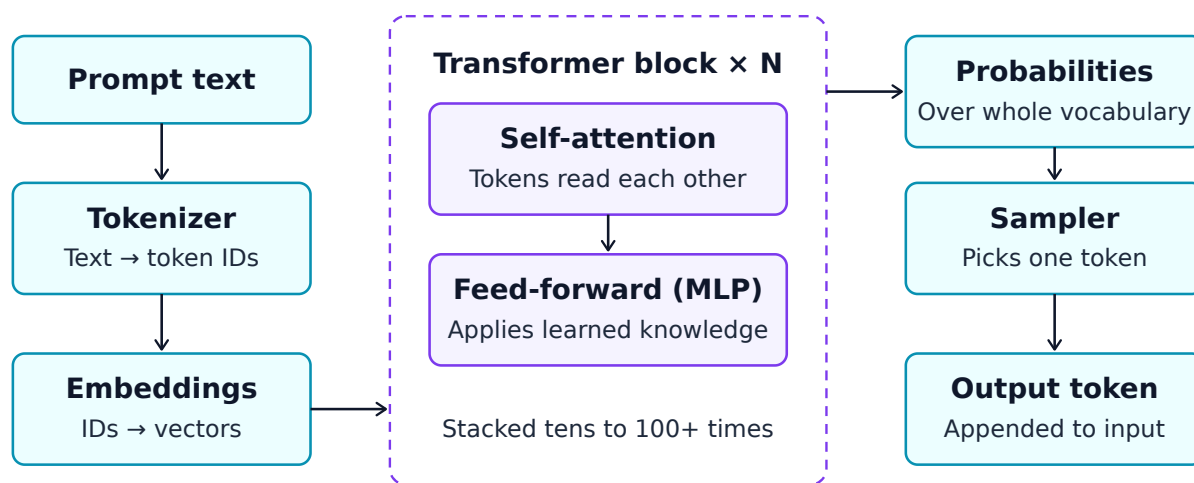
Figure 1. The build phase (blue) ends in a frozen set of weights. The implement phase (green) runs those weights for every request.

What each build stage contributes

- **Data.** Trillions of tokens of text, code, images and licensed or synthetic material. It is deduplicated, quality-filtered, and converted to tokens, the sub-word integer IDs the model actually sees. Data quality is now as much a differentiator as quantity.
- **Pretraining.** A single objective: predict the next token. Gradient descent adjusts the parameters until prediction error stops falling. This is where nearly all the raw knowledge and the bulk of the compute goes. The output is a "base model" that completes text but does not follow instructions.
- **Post-training.** Supervised fine-tuning (SFT) on example dialogues, then reinforcement learning, either from human preference (RLHF) or from verifiable outcomes such as tests that pass or math that checks. This turns a text completer into an assistant and produces "reasoning" behavior. Distillation, training a smaller model on a larger model's outputs, also sits here.
- **Evaluation.** Benchmarks, red-teaming and safety testing gate release. A pass freezes the weights.

2. Inside the model: one pass per token

The frozen weights implement a **transformer**. The model does not look anything up or run a program in the usual sense. It converts text to numbers, pushes those numbers through a deep stack of identical layers, and produces a probability for every possible next token. One token is chosen, appended to the input, and the whole process runs again.



↻ One full pass through all N blocks per output token, then repeat

Figure 2. Input and output handling (cyan) surrounds the transformer stack (purple). Generation is a loop: every output token costs one full pass.

Three consequences of the loop

- **Input is cheap, output is expensive.** The whole prompt is processed in one parallel pass, called prefill. Each output token needs its own sequential pass, called decode. This is the mechanical reason output tokens are priced several times higher than input tokens, and why total token count tends to drive cost-per-task more than list price does.
- **The KV cache is the memory tax.** Attention stores intermediate results for every prior token so they are not recomputed. That cache grows with context length and consumes accelerator memory, which makes serving bound by memory bandwidth more than by arithmetic. Cached-prefix discounts exist because a provider can reuse this state.

- **"Reasoning" is more tokens through the same loop.** Effort tiers buy additional decode passes before the answer. The network itself does not change.

3. Scale: why building is a capital project

Capability has tracked training compute closely, so labs have scaled it relentlessly. Epoch AI's trend dashboard puts frontier language-model training compute at about 5× per year since 2020, a doubling roughly every 5.2 months, with training costs rising about 3.5× per year and power requirements doubling annually. **HIGH**

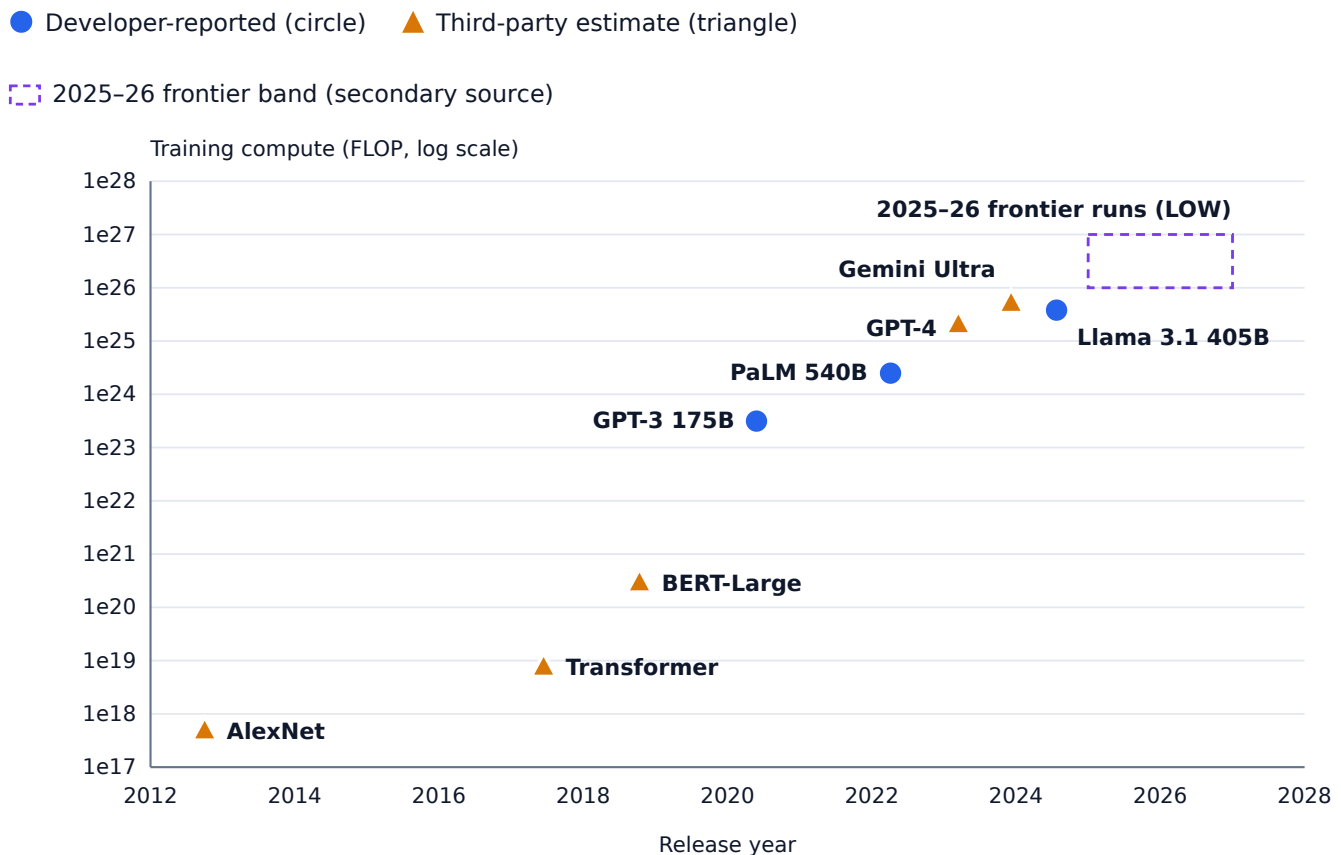


Figure 3. About eight orders of magnitude in twelve years. Marker shape encodes provenance: circles are developer-reported, triangles are third-party estimates. Hover a marker for its value. The 2025–26 band is drawn as a range, not as points, because it is not lab-disclosed.

Epoch's own estimates put GPT-4 at about 2e25 FLOP and Gemini Ultra at about 5e25, against GPT-3's roughly 3e23 in 2020. **HIGH** A secondary analysis places 2026 frontier runs in the 1e26 to 1e27 window. **LOW**

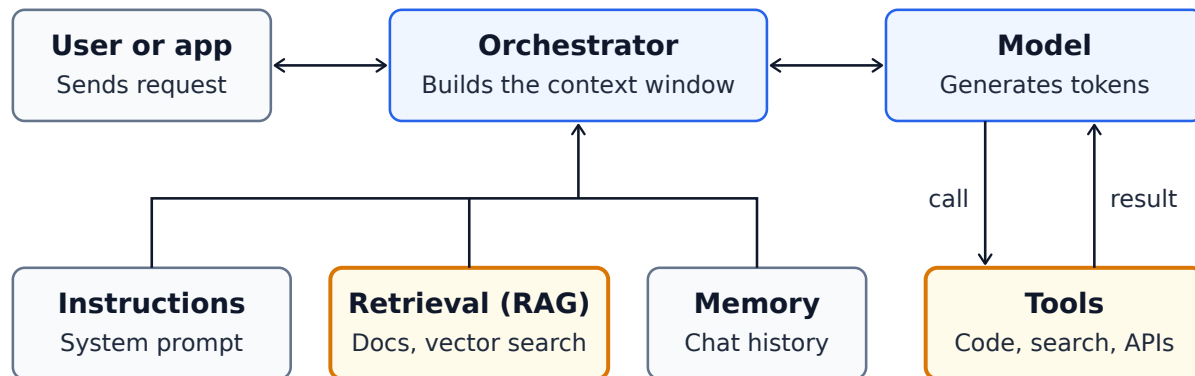
The physical stack, bottom to top

LAYER	WHAT IT IS	TYPICAL BINDING CONSTRAINT
Power and datacenter	Gigawatt-scale sites, cooling, grid interconnect	Energy availability, build time
Accelerators	GPUs and TPUs with high-bandwidth memory (HBM)	Chip and HBM supply, advanced packaging
Interconnect	NVLink, InfiniBand or Ethernet fabrics linking tens of thousands of chips	All-to-all bandwidth during training
Systems software	CUDA / ROCm, PyTorch or JAX, parallelism frameworks	Utilization, fault recovery at scale
Model	Architecture plus trained weights	Data quality, algorithmic efficiency
Serving	vLLM, TensorRT-LLM, llama.cpp, quantization	Memory bandwidth, KV cache size
Orchestration and app	Context assembly, tools, agents, user interface	Reliability, security, cost per task

The "binding constraint" column is synthesis, not a sourced ranking. MED

4. Implementation: the model is one component

A deployed AI product is mostly conventional software wrapped around the model call. The model is stateless and knows only what is in its **context window** on that pass. Everything that feels like knowledge of your files, memory of past conversations, or the ability to act is the orchestration layer putting text into that window and executing what comes out.



🟡 Amber = where untrusted external content enters the context

🔄 Agent loop: tool result re-enters the context and the model runs again until done

Figure 4. The orchestrator assembles instructions, retrieved documents and history into one context window. Tools close the loop that makes a model an agent.

Implementation patterns, in increasing order of autonomy and risk

- **Prompting.** Instructions plus the request. It is the cheapest lever and is often underused.
- **Retrieval-augmented generation (RAG).** Relevant documents are fetched and pasted into the context, so answers are grounded in current or private data without retraining.
- **Fine-tuning.** Small additional training, such as LoRA adapters. It changes format, tone or a narrow skill, and is a poor fit for injecting facts.
- **Tool use.** The model emits a structured call, the orchestrator executes it and returns the result. The Model Context Protocol (MCP) standardizes this interface.
- **Agents.** Tool use in a loop, with the model deciding the next step. Per-step errors compound and cost scales with loop length. If each step succeeds with probability p , a chain of N independent steps succeeds at most p^N of the time, which is an upper bound on decay rather than a forecast.

Security note: the lethal trifecta. An agent becomes dangerous when three things coincide: access to private data, exposure to untrusted content, and the ability to act autonomously. In Figure 4, retrieval and tool results are the entry points for untrusted content, and tools are also the action surface. The risk therefore concentrates in the orchestration layer, not in the model.

Where the weights run: three deployment modes

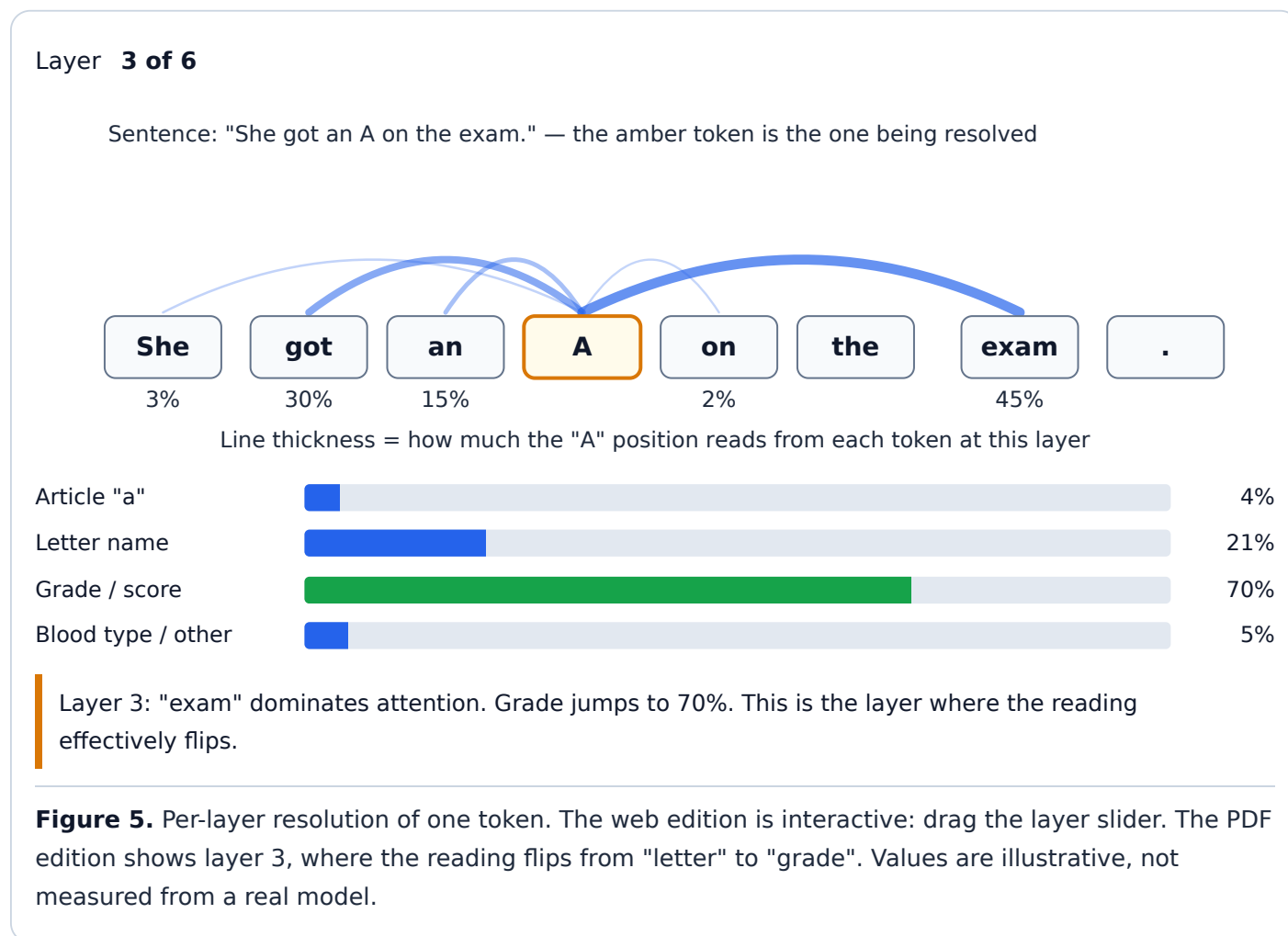
MODE	WHO HOLDS THE WEIGHTS	COST SHAPE	TRADE-OFF	ROUTING LANE
Local open weights	You (llama.cpp, Ollama, quantized to fit consumer VRAM)	Hardware capex, near-zero marginal cost	Private and free per token; capability capped by memory	Lane 1
Hosted open weights	A cloud host behind a router	Low per-token opex	Cheap and capable; endpoint reliability varies	Lane 2
Closed frontier API	The lab only	Highest per-token opex	Top capability; no weight access	Lane 3

The routing decision across these modes is an intelligence-per-dollar question: cost per unit of completed task, including endpoint reliability, rather than headline capability.

6. How a model reads one token, and where backpropagation fits

A common mental model is that a language model "reads" a character such as **A**, tries several interpretations, and backtracks when one fails. The real mechanism is different in two ways. First, the model never sees a shape: the tokenizer hands it an integer ID, and the work is deciding what that ID means *here* (the article "a", the letter's name, a grade, a blood type, a musical note, a variable). Second, within a single forward pass nothing backtracks. Each layer rewrites the token's vector, so an early, generic reading is overwritten by a context-specific one. Interpretability researchers call reading those

intermediate states out the **logit lens**; the process itself is **iterative refinement in the residual stream**.



Two things to notice. The "article" reading at layer 0 was never wrong: it was the correct prior with no context. And the flip at layer 3 is not a decision the model could revisit; it is arithmetic. Attention added the "exam" vector into the "A" position, and the sum now points in the "grade" direction.

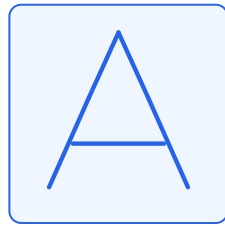
Recognizing A as a *shape* is the job of a vision encoder in a multimodal model. It is also a single forward pass, but the hierarchy is spatial rather than contextual, and it is the one place where the picture of competing hypotheses is closest to literal.

1. Pixels → patches



Each cell → one vector

2. Edges



Orientation, contrast

3. Strokes

Left diagonal ✓

Right diagonal ✓

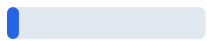
Crossbar ✓

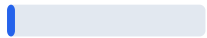
Apex join ✓

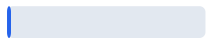
Mid-layer features

4. Hypotheses

A 

H 

Λ 

4 

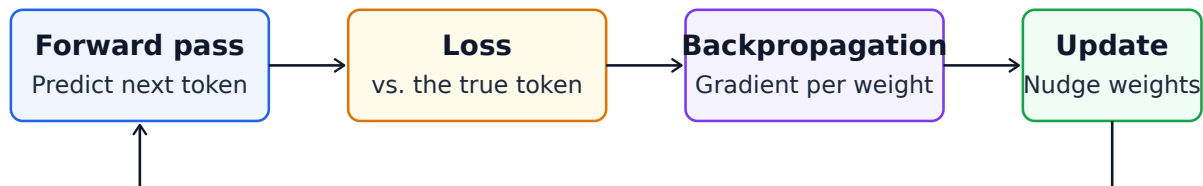
Crossbar kills Λ, apex kills H

Figure 6. Spatial hierarchy in a vision encoder. Λ and H are not tried and abandoned; their evidence is simply outweighed in the same forward sum.

The correct terms

WHAT IS HAPPENING	TERM	WHERE IT RUNS
Token meaning refined layer by layer	Iterative refinement in the residual stream; read out with the logit lens or tuned lens	Inference, inside one forward pass
Competing readings coexisting until one dominates	Superposition of features, resolved by contextualization	Inference, inside one forward pass
Model writes a hypothesis, then corrects it	Self-correction, self-verification, reflection	Inference, in the output tokens
Several candidate continuations kept, losers dropped	Beam search; search over reasoning paths (tree of thoughts, best-of-N)	Inference, across candidate outputs
Error pushed back through the layers to adjust weights	Backpropagation , with gradient descent as the update rule	Training only

So the process that resembles backtracking through the layers is **backpropagation**, and it happens only during training. After a wrong prediction, the backward pass computes how much each weight contributed to the error, layer by layer from the output back to the input, and gradient descent nudges each one. It is how the stroke detectors in Figure 6 were learned. Once the weights are frozen for deployment it never runs again.



↻ Repeated for trillions of tokens; the loop stops when the weights are frozen for release

Cost: the backward side is about 2× the forward side, so a training token costs $\approx 3\times$ an inference token, and needs full-precision weights plus optimizer state in memory. Inference can run quantized on frozen weights.

Figure 7. Backpropagation is the training-time loop. Its roughly 3× per-token cost and memory footprint are the mechanical root of the capex/opex split in Figure 1.

7. Provenance and confidence

- **HIGH** Lifecycle structure, transformer loop, prefill/decode asymmetry, KV cache role, implementation patterns. Stable, widely documented engineering.
- **HIGH** The 5× per year compute trend and the GPT-3, GPT-4 and Gemini Ultra compute figures, checked against Epoch AI pages on 19 Sep 2026. The GPT-4 and Gemini Ultra figures are Epoch estimates, not lab disclosures.
- **MED** The AlexNet, Transformer, BERT-Large, PaLM and Llama 3.1 405B chart points are from recall of the source papers and Epoch's dataset and were not re-pulled for this revision. PaLM and Llama are developer-reported values; the other three are third-party estimates.
- **MED** The stack table's "binding constraint" column is the author's synthesis.
- **HIGH** Section 6 mechanism: one-directional forward pass, residual-stream refinement, logit lens as a read-out, and the placement of backpropagation, self-correction and search. Standard usage in the ML and interpretability literature.
- **MED** The $\approx 3\times$ training-to-inference cost per token is the standard rule of thumb (forward $\approx 2 \cdot N \cdot D$ FLOP, full step $\approx 6 \cdot N \cdot D$), not a measured figure for any named model.
- **LOW** The per-layer percentages and attention weights in Figure 5 and the hypothesis bars in Figure 6 are illustrative constructions, not measurements.

- **LOW** The 2025–26 band of 1e26 to 1e27 FLOP comes from a single secondary consultancy write-up citing Epoch, not from lab disclosure.

Sources

- Epoch AI — Trends in Artificial Intelligence (compute growth rate, doubling time, cost and power growth)
- Epoch AI — Training compute of frontier AI models grows by 4–5× per year (GPT-3, GPT-4, Gemini Ultra estimates)
- Epoch AI — Data on AI Models (notable models dataset)
- Deluair Consultancy — Frontier AI training cost trajectory 2026 (secondary source for the 2025–26 band)

How AI Is Built and Implemented · REV 1.1 · 19 September 2026 · Vendor-reported and third-party figures are kept distinct throughout.